higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATERIAL TECHNOLOGY
NQF LEVEL 3

4 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 D
1.2 H
1.3 J
1.4 E
1.5 C
1.6 F
1.7 G
1.8 B

(8 × 1) **[8]**

QUESTION 2

2.1 True
2.2 False
2.3 False
2.4 False
2.5 True
2.6 True
2.7 True
2.8 False

(8 × 1) **[8]**

QUESTION 3

3.1 3.1.1 Mass per unit volume

3.1.2 Rate at which heat can flow through a metal

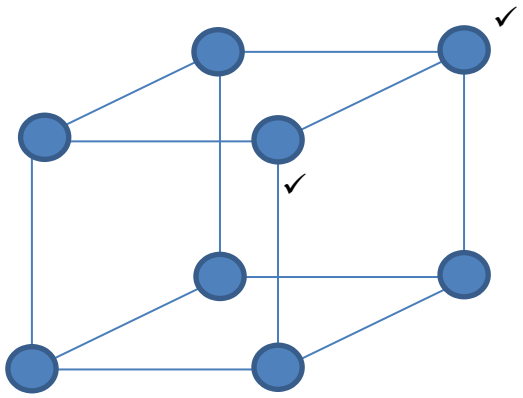
3.1.3 Relates to the quality of light reflected from a metal

3.1.4 Temperature at which a metal changes from liquid state to a vapour state

(4 × 2) (8)

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3.2



1 mark for all atoms in corner positions
1 mark for atom in centre of cube
1 mark if a total of 9 atoms are indicated in correct positions (3)

3.3 3.3.1 Cu
3.3.2 Mn
3.3.3 Pb
3.3.4 He
3.3.5 Al

(5 × 1) (5)

3.4 Improves the strength
Increases the ductility
Improves the toughness
Improves resistance to corrosion
Increases impact resistance
Improves the ability to resist fatigue (Any 5 × 1) (5)
[21]

QUESTION 4

4.1 4.1.1 B
4.1.2 B
4.1.3 B
4.1.4 A
4.1.5 A

(5 × 1) (5)

4.2 Properties
- High tensile strength
- High compressive strength
- High strength/weight ratio
- Dimensional stability
- Easy to fabricate
- Resistance to corrosion

Uses
- Boats
- Trailers
- Aircraft
- Household appliances
- Water and fuel tanks
- Surfboards

Vehicle components (Any 4 × 2) (8)

4.3 4.3.1 Substances which have a molecular structure built up from a large number of similar units bonded together

4.3.2 Molecules that can be bonded to other identical molecules in long repeating chains to form a polymer

4.3.3 Type of synthetic or natural polymer that has elastic properties (3 × 2) (6)

4.4
- Low strength
- Poor weather resistance
- Poor heat resistance
- Environmental hazard

Low dimensional stability (Any 2 × 1) (2)
[21]

QUESTION 5

5.1
- One of the few methods of continuous melting
- Relatively low operating costs
- Easy cleaning and repairing, as bottom can drop away
- High melt rates

Ease of operation (Any 4 × 1) (4)

5.2 Hot compressed air blown into the converter from the bottom.✓✓ The oxygen in the air reacts with the iron to produce iron-oxide.✓✓ The iron-oxide reacts with the impurities in the metal, and a layer of slag is formed. The slag is then removed, and the iron is poured into moulds.✓✓ (Any 3 × 2) (6)

5.3 The hardened steel is heated to a temperature in the austenitic region.✓✓ The steel is kept at this temperature until austenite is formed.✓✓ The steel is then slowly cooled down in a furnace at the rate of about 20 °C per hour, so that the austenite turns back into ferrite and pearlite.✓✓ (3 × 2) (6)

5.4 Yes (1)

5.5 Oil (1)

5.6
- Pearlite
- Austenite
- Critical point
- Martensile

(Any 3 × 1) (3)
[21]

QUESTION 6

6.1 6.1.1 circular
6.1.2 quality
6.1.3 hardness
(3 × 1) (3)

6.2 Mark or recess on the surface of a material✓ once a hardness test has been performed on the material✓ (2)

6.3 6.3.1 $\Delta L = L_f - L_o$

$= 185.98 - 170$ ✓

$= 15{,}98mm$ ✓ (3)

6.3.2 No unit (1)

6.3.3 $\varepsilon = \frac{\Delta L}{L}$

$= \frac{15{,}98}{170}$ ✓

$= 0{,}094$ ✓ (2)

6.3.4 $8\ mm = 0{,}008\ m$ ✓

$A = \frac{\pi}{4} d^2$ ✓

$= \frac{\pi}{4}(0{,}008)^2$

$= 0{,}00005\ m^2$ ✓ (3)

6.3.5 $\sigma_{max} = \frac{F_{max}}{A}$

$= \frac{25000}{0{,}00005}$ ✓

$= 500MPa$ ✓ (2)

6.4 6.4.1 $Surface\ area\ of\ impression = \frac{\pi D}{2}\left[D - \sqrt{D^2 - d^2}\right]$

$= \frac{\pi \times 10}{2}\left[10 - \sqrt{10^2 - 4{,}54^2}\right]$ ✓

$= 17{,}12mm^2$ ✓ (2)

6.4.2 $BHN = \frac{Load\ in\ kg}{Surface\ area\ of\ impression}$

$\therefore Load = BHN \times Surface\ area\ of\ impression$ ✓

$= 204{,}44 \times 17{,}12$ ✓

$= 3500kg$ ✓ (3)

[21]

TOTAL: 100

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